



MXenes: Types, preparation, and applications in water treatment

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Abstract

MXenes have attracted significant attention as advanced 2D materials for water treatment, but previous assessments have been largely descriptive and have not adequately compared material types, manufacturing strategies, and practical performance. This paper aims to evaluate MXene-based membranes in terms of composition, surface chemistry, interlayer engineering and membrane design and to relate properties to water flux, rejection efficiency, adsorption capacity, antifouling behaviour and stability. Our goal is not to provide general summaries, but to highlight the critical points that will guide MXenes from laboratory demonstrations to real-world applications in water purification. This review also addresses major challenges including oxidation, restacking, swelling, mechanical fragility, scalability and environmental safety. Such a contrast between existing gaps and present advances provides a more lucid basis for the selection and design of MXene membranes in sustainable water treatment applications.

Keywords: MXene; $Ti_3C_2T_x$; membranes; water treatment; nanofiltration.

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1- Introduction

In the last 50 years, industrial development and urbanisation have been major drivers of a marked increase in global water use [1]. Water pollution and freshwater scarcity are major global issues, particularly in regions affected by industrial discharge, salinization, and population growth [2]. Conventional treatment technologies have difficulties in treating complex pollutant mixtures, trace pollutants, and the trade-off between flow and selectivity [3]. Advanced nanomaterials have been investigated for incorporation into adsorbents and membranes for the purpose of enhancing water purification efficiency. The MXenes are of particular interest due to their tunable chemistry in the production and structural properties that allow controlled transport and selective sorption [4–7].

MXenes constitute a class of two-dimensional materials primarily originating from stacked MAX phases [8, 9]. Since their discovery, MXenes have suitable characteristics to control volume expansion, charge storage capability, and potential for several energy storage applications such as batteries and supercapacitors [4, 10]. The advantages of MXenes are superior to other materials, such as larger interlayer spacing, remarkable electrical conductivity, rapid diffusion of ions, easily accessible composition, better thermal stability, hydrophilic nature, and large surface area [11].

In water treatment, they hold significant importance as they can function either as standalone materials or as components of composite membranes [12, 13]. Due to

their capacity to create stable films, lamellar membranes, and hybrid structures, they have garnered extensive study attention [14]. This review is innovative in the comparative and critical review of the MXene membrane design for water treatment, focusing on the synthesis-dependent performance, stability challenges, and translational barriers that are not comprehensively unified in recent reviews. The review summarises previous work and provides a critical comparison of MXene compositions, fabrication routes, and practical limitations. To better understand the scope of the present review, Table 1 compares it with recent MXene reviews published in the 2022–2026 period, highlighting the specific areas in which our manuscript provides a more critical and water-treatment-oriented analysis.

2- MXene family and classification

MXenes are a group of layered transition-metal carbides, nitrides, and carbonitrides that are typically obtained by removing the A-layer from MAX phases. In the parent MAX phase, M is an early transition metal such as *Ti*, *V*, *Nb*, or *Mo*; A is usually a group 13 or 14 element such as *Al* or *Si*; and X is carbon, nitrogen, or both [15].



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Table 1. MXene reviews published between 2022 and 2026

Review / Year	Main focus	Water-treatment scope	Membrane fabrication/design	Critical comparison of MXene types	Practical performance discussion	Gaps/limitations addressed	Distinct contribution	Ref.
Two-dimensional MXenes: recent emerging applications (2022)	Broad emerging applications of MXenes.	Limited and general, not focused on water treatment alone.	Not a central emphasis.	General overview, not a detailed materials comparison.	Limited critical analysis of real application constraints.	Basic challenges discussed, but not deeply tied to water treatment deployment.	Early broad MXene overview with cross-disciplinary scope.	[21]
Advances of MXenes; Perspectives on Biomedical Research (2022)	Biomedical applications of MXenes.	Not a water-treatment review.	Not relevant.	Discusses functional MXenes, but for biomedical use.	Performance is framed for biomedical systems, not membranes.	Biomedical challenges only.	Demonstrates how MXenes are adapted for a different application field.	[22]
Latest advances and comparative analysis of MXenes as anode and cathode electrodes (2023)	Electrochemical energy storage.	Not water-treatment focused.	Not relevant.	Compares MXene roles as electrodes.	Performance is electrochemical, not separation-related.	Battery-specific issues.	Useful for comparative methodology, but not relevant to water treatment.	[23]
Application of MXenes for water treatment and energy-efficient desalination: A review (2022)	Water treatment and desalination.	Yes, but with emphasis on desalination and broader treatment routes.	Discusses MXenes in membranes and CDI.	General MXene-composite discussion.	Includes kinetics, isotherms, and operating conditions.	Addresses regeneration and long-term use, but more application-driven than membrane-design-focused.	Strong foundation for desalination and adsorption, but not deeply comparative across membrane fabrication strategies.	[24]
Recent advances in applications of MXenes for desalination, water purification and as an antibacterial: a review (2024)	Water purification, desalination, antibacterial action.	Yes.	Discusses membrane and antibacterial uses.	General comparison of MXene performance.	Notes some durability and purification efficiency.	Limited by broad scope and limited fabrication-critical discussion.	Useful overview, but still mostly descriptive.	[25]
Advancements in Water Treatment with MXene-Enhanced Membranes: A Review of Current Progress and Future Directions (2024)	MXene-enhanced membranes.	Strong focus.	Yes.	Discusses synthesis, fabrication, application.	Reports high flux and rejection, durability, and comparative membrane performance.	Still largely a literature synthesis rather than a structured gap-analysis across MXene classes and methods.	Stronger practical emphasis on membrane durability and separation efficiency.	[26]
MXene-based materials for enhanced water quality: Advances in remediation strategies (2025)	Remediation strategies for water quality.	Yes.	Limited membrane-specific depth.	Reviews MXene materials broadly.	Discusses remediation performance.	Not primarily a membrane-fabrication review.	Broader remediation perspective than membrane engineering.	[27]
MXenes in membrane engineering: Unlocking the new potential for water treatment (2025)	MXenes in membrane engineering.	Yes.	Yes, central theme.	Addresses engineered membranes and modification.	Focuses on practical membrane improvements.	Some scope overlap with your manuscript, but not necessarily with your exact comparative structure.	Strong membrane-engineering perspective.	[28]
Recent progress and prospect of MXene-based nanofiltration membranes (2025)	Nanofiltration membranes.	Yes, but limited to NF.	Yes.	Likely focused on NF-oriented materials and design.	Performance is assessed around NF metrics.	Narrower scope than your manuscript.	Excellent for nanofiltration-specific context.	[17]
A comprehensive review on MXenes: synthesis, stability, properties and functionalization (2026)	General MXene synthesis and functionalization.	Not water-treatment focused.	Not central.	Broad MXene comparison.	Not membrane-specific.	General challenges and opportunities.	Useful background, but not a water-treatment review.	[29]
A comprehensive overview of recent progress in MXene-based ... (2024)	General MXene progress.	Limited or indirect.	Not necessarily central.	Broad overview.	Broad application discussion.	Limited membrane-specific and water-treatment-specific critical depth.	General state-of-the-art MXene overview.	[30]

The resulting MXene retains a layered structure, but gains surface terminations from the etching and post-

treatment process. General formula $M_{n+1}X_nT_x$, defined by “ $n + 1$ ” transition metal layers of M element comprising

“ n ” layers of ‘ X ,’ where it can be a carbon or nitrogen element. X can also include oxygen after the recent synthesis of oxycarbide MXenes (substitution of oxygen in carbide MXenes [5,16–19]. T_x represents surface terminations where elements of groups 15, 16, and 17 of the periodic table and hydroxyl groups can be attached to MXene during the etching process, and x in T_x can be less than or equal to 2 as shown in Fig. 1 [20].

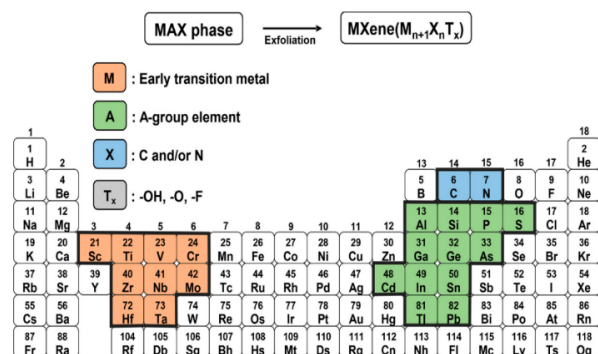


Fig. 1. Periodic table showing the composition of the MAX phase and MXene, reproduced from [20], licensed under CC BY 4.0

The most researched MXene is $Ti_3C_2T_x$, where T_x is a mixture of surface groups, usually fluorine, hydroxyl, and oxygen [31]. Because of its hydrophilicity, negative surface charge, and strong interaction with polar molecules and ions, this composition is highly relevant for water-treatment applications [32]. There have also been reports of other MXenes, such as Ti_2C , V_2C , Nb_2C , and Mo_2C , which may exhibit distinct transport or adsorption behaviour based on their electronic structure and surface chemistry [7]. Carbonitrides (MXenes) are less common in water treatment than carbide-based systems [33].

The classification can also be evaluated in terms of surface termination, chemical complexity and layer thickness. Some MXenes are mainly terminated with O or OH groups, while some have high fluorine content depending on the etching method used [34]. As such, these surface groups are known to influence colloidal stability, membrane formation and binding affinity for dissolved pollutants. So, the “type” of MXene is connected to its composition, surface chemistry, and processing history [35, 36].

Table 2 summarises the relative performance of the different MXene membrane platforms in terms of water flux, rejection efficiency, adsorption capacity, antifouling behaviour, stability, advantages and limitations to better distinguish between the practical strengths and limitations.

3- Preparation of MXenes

MXenes are generally produced through the selective etching of the A-element from MAX phases. The predominant technique involves the utilisation of hydrofluoric acid or fluoride-based etching techniques that eliminate the A-layer, resulting in multilayer MXene stacks [34]. Although effective, these approaches also

present safety and environmental issues, prompting the creation of safer alternatives. The resulting product is typically subjected to repeated washing, centrifugation, and neutralisation before additional processing [42]. This process selectively etches the Al layer while preserving the $Ti - C$ bonds, as the $Ti - A$ bonds are stronger in comparison, as shown in Fig. 2 [43]. As a result, surface groups such as F , OH , or a combination of both are formed and attached to the MXene surface [44].

Following etching, the multilayer structure is typically intercalated with molecules or ions that augment the space between the sheets. Typical intercalants include dimethyl sulfoxide, tetraalkylammonium compounds, lithium ions, and other small molecules. This phase diminishes the tensions between the layers, facilitating their separation into single or few-layer nanosheets. Delaminated MXenes are particularly appealing as they optimise the available surface area and facilitate the processing into thin films and membranes [16].

Alternative synthesis methods are attracting interest due to the constraints of fluoride-based etching. These encompass electrochemical etching [45], molten-salt techniques [46], and alkali-assisted methodologies [47], as depicted in Fig. 3 [48]. These approaches may enhance safety, diminish faults, and provide more regulated surface chemistry. Nonetheless, they may also present novel issues concerning yield, repeatability, or scalability. The optimal synthesis pathway for water treatment should yield MXenes that are stable, delaminated, and appropriate for large-scale membrane fabrication [49].

The synthesis route is critical to structural quality, surface chemistry, cost, scalability, safety and overall utility of MXenes in water treatment applications [41,50]. The most well-known and common technique is fluoride-based etching, that has relatively high yields and practicality, but poses safety problems and environmental problems [51]. Conversely, novel techniques such as electrochemical etching [9], molten-salt processing [52] and alkali-assisted methods [53] provide safer and potentially more sustainable alternatives, but come with trade-offs in process complexity, reproducibility, yield or scalability. Thus the selection of a preparation method should be based on performance, operational safety, economic feasibility and manufacturability in principle, especially when the application target is the preparation of MXene-based membranes for large-scale water purification [50, 53].

4- Surface chemistry and properties

The surface chemistry is very important for the attractiveness of MXenes for water treatment applications. During synthesis, new surfaces are terminated with functional groups such as $-O$, $-OH$ and $-F$ [54]. These groups have a strong effect on the wettability, the interactions between layers and the adsorption behaviour. Proper functional groups can significantly enhance MXene hydrophilicity, thus greatly enhancing membrane wetting and water transportation.

They also affect the surface charge, which is important for the rejection or binding of dissolved ions and dyes [55].

Another interesting feature is its two-dimensional layered structure. The sheets are stacked one on top of the other and create nanochannels in between, which help to segregate molecules and ions. These channels can be altered by intercalation, cross-linking or hybridisation with polymers and other nanomaterials. Hence, MXenes can serve as an active component rather than a mere filler in water-treatment membranes [56].

The key challenges of MXene materials in wastewater treatment are structural instability, oxidation sensitivity, membrane swelling and limited long-term reusability. Different functionalisation procedures, including surface chemical modification, doping, intercalation and composite synthesis, have been developed as shown in Fig. 4 [57]. The strategies are aimed at improving the adsorption capacity, separation efficiency, structural

stability, and general flexibility of MXene-based materials in wastewater treatment [28]. The chemical modification of the surface allows the tuning of functional groups on the surface of MXene, which improves the interactions with pollutants and ion transport, binding of contaminants, and antifouling properties. Interlayer spacing and surface electrochemical properties of MXenes can be tuned by doping and intercalation, which enhances permeability, selectivity and affinity for particular pollutants [57]. Adding MXenes to composites with polymers, carbon materials, metal oxides or other functional nanomaterials can greatly improve the mechanical strength, chemical stability and treatment efficiency. These functionalisation procedures are promising approaches to improve the properties of MXene-based materials and their application in advanced wastewater treatment technologies [38, 58, 59].

Table 2. relative performance in terms of water flux, rejection efficiency, adsorption capacity, antifouling behaviour, stability, advantages, and limitations

MXene membrane type	Water flux	Rejection efficiency	Adsorption capacity	Antifouling performance	Stability	Advantages	Limitations	Ref.
Pure Ti_3C_2Tx lamellar membrane	Typically high at the beginning of operation because of hydrophilic surfaces and open nanochannels, but flux may decrease during use because of swelling and restacking.	High rejection for dyes and charged contaminants when interlayer spacing is properly controlled.	High affinity for cationic dyes and some metal ions due to abundant surface terminations and negative surface charge.	Moderate; hydrophilicity reduces some fouling, but organic fouling and pore blockage remain concerns.	Limited by oxidation, swelling, and structural restacking.	Easy to prepare, widely studied, and serves as the main benchmark in the literature.	Poor long-term robustness and limited mechanical strength as a standalone membrane.	[37]
Ti_3C_2Tx MXene-polymer composite membrane	Often more stable than pure MXene membranes because the polymer matrix improves structure and processability.	High rejection can be retained, although performance depends strongly on MXene dispersion and loading.	Good adsorption performance, but some active sites may be partially masked by the polymer phase.	Better than pure MXene membranes in many cases because the polymer support improves surface integrity and reduces fouling deposition.	Better mechanical and operational stability than free-standing MXene films.	More suitable for practical membrane fabrication and scale-up.	Aggregation, interfacial defects, and no uniform dispersion can reduce performance.	[38]
Crosslinked MXene membrane	Usually slightly lower than loose lamellar membranes at first, but more stable during long-term filtration.	Improved selectivity because crosslinking stabilizes the transport channels and limits swelling.	Moderate to high, depending on whether functional sites remain accessible after crosslinking.	Good antifouling behaviour because channel collapse and excessive swelling are suppressed.	Stronger resistance to water-induced swelling and layer restacking.	Better durability and channel stability.	Excessive crosslinking may reduce permeability and slow transport.	[39]
Intercalated MXene membrane	Can be high when intercalation enlarges and stabilizes the interlayer galleries.	Often improved because interlayer spacing becomes more controlled.	Enhanced when the intercalant contributes additional binding sites or charge interactions.	Improved relative to untreated membranes because swelling and compaction are reduced.	Better structural stability than pristine films, although this depends on intercalant retention.	Tunable channel size and improved flux/selectivity balance.	Some intercalants may leach or lose effectiveness over time.	[40]
Hybrid MXene membrane with GO, CNTs, metal oxides, or related fillers	Often high and more stable because the hybrid structure prevents collapse and improves water transport pathways.	Very good for dyes, salts, and mixed contaminants when the hybrid architecture is optimized.	Can be high if the secondary filler provides extra adsorption or catalytic sites.	Strong antifouling performance in many cases because of better hydrophilicity, spacing control, and reduced aggregation.	Usually better than pure MXene membranes because the hybrid phase improves oxidation resistance and mechanical integrity.	Provides the best balance between permeability, selectivity, and durability in many reports.	Fabrication is more complex and requires careful optimization to avoid heterogeneity.	[35]
Conductive MXene membrane under electrochemical assistance	Flux can be improved because electrochemical operation helps suppress fouling and support cleaning.	Strong rejection with added control over separation behaviour.	Usually secondary to separation, although conductivity may support reactive treatment pathways.	Very good antifouling performance because electrochemical assistance can reduce deposition and aid in cleaning.	Better during operation, but the system requires electrode integration and operational control.	Useful for dynamic fouling control and advanced treatment platforms.	More complex module design and energy demand are required.	[41]

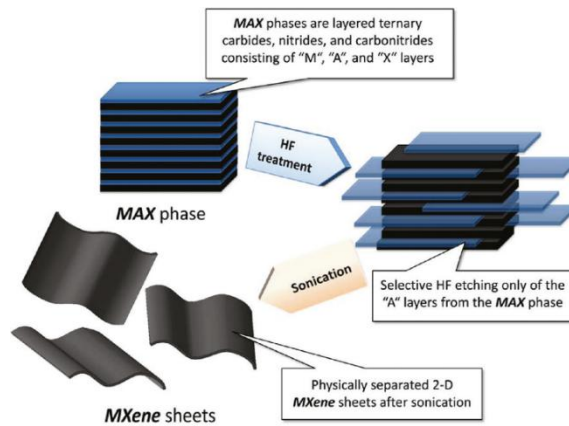


Fig. 2. Basic map of etching process of the MAX phase to MXene nanosheets, reproduced from [43], licensed under CC BY 4.0

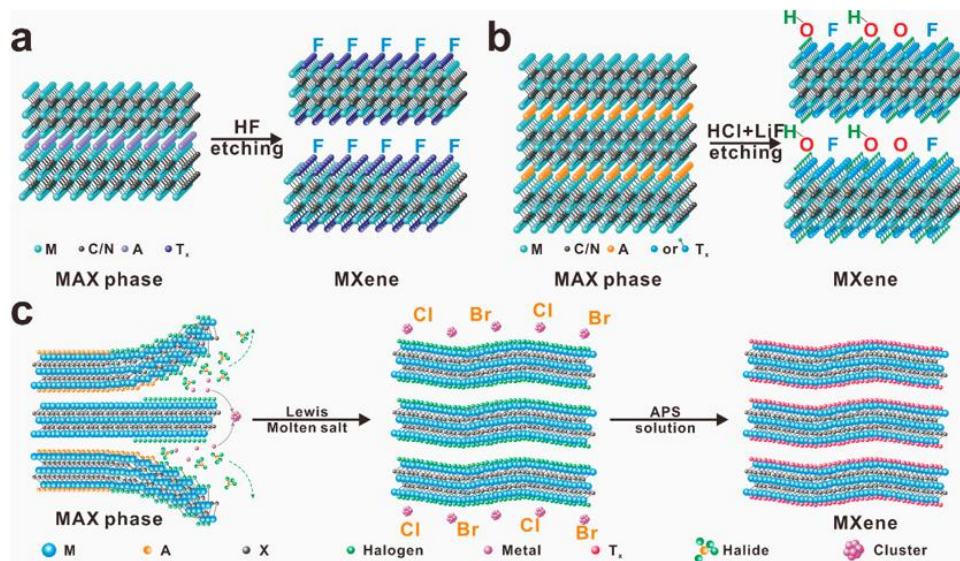


Fig. 3. MXene synthesis by selective etching MAX phase : (a) HF-based acidic etching method, resulting in abundant fluorinated groups ($-F$); (b) fluoride salt based acidic etching method, leading to the distribution of various surface groups including $=O$, $-OH$, $-F$; and (c) Lewis molten salt etching method for generation of halogen group ($-Cl$ or Br), reproduced from [48], licensed under CC BY 4.0

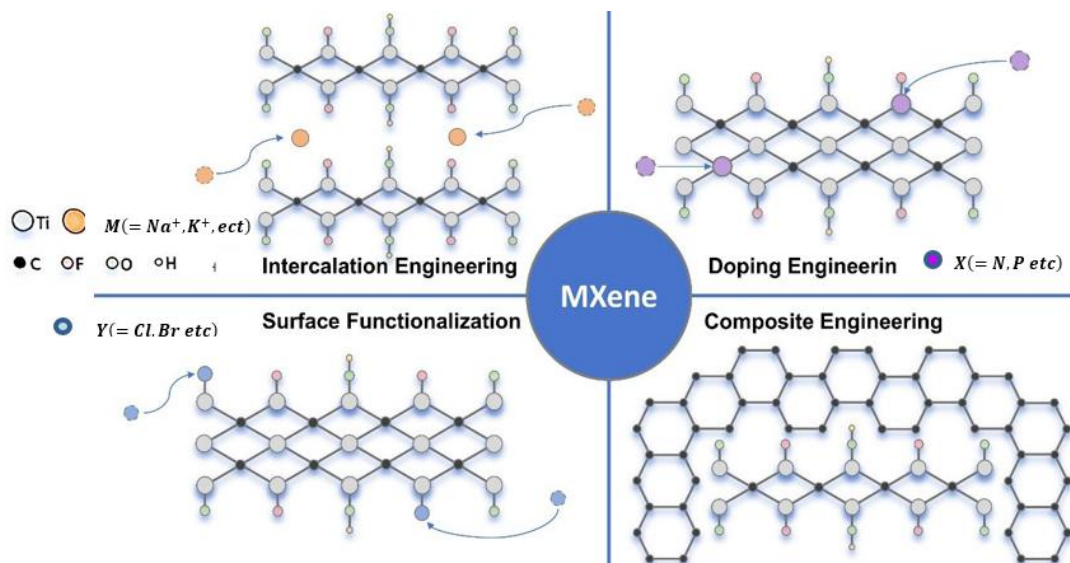


Fig. 4. Schematic diagram of multi-engineering strategies for MXene structural optimization, reproduced from [57], licensed under CC BY 4.0

5- MXene membrane fabrication

MXenes can be turned into membranes in many ways. The most widely used method is vacuum filtering [60]. This technology allows for the generation of dense or semi-dense films with designed thicknesses and the organised stacking of thin nanosheets. This gives a membrane structure which is especially suited to size and charge-based separation. Vacuum filtering is more advantageous for laboratory-scale processing [50,61] in contrast to the industrial-scale throughput.

Other fabrication methods are coating [62], spray deposition [63], layer-by-layer assembly [64], and inclusion into polymer matrices. Mixed-matrix membranes based on MXenes have great advantages since they combine the separation efficiency of MXenes with the mechanical strength and processability of polymers. Polymer supports most frequently employed are Polyethersulfone (PESU) [65], Polyacrylonitrile (PAN) [66], Polyvinylidene fluoride (PVDF) [67], and Polyamide systems. MXene can improve the hydrophilicity, anti-fouling, and transport channels of these composites [40, 68].

One of the main objectives in membrane fabrication is to prevent restacking and to ensure the stability of nanochannels [69]. To this end, researchers employ intercalating pillars, crosslinkers and hybrid fillers such as graphene oxide, metal oxides and biopolymers. These improvements make it easier to save space when in use and to prevent collapse or compacting. In water treatment applications the morphology of the membrane is often as important as its chemical composition [50, 70]. The fabrication of MXene-based membranes is modulated by key factors including the processing parameters [71], membrane morphology [72], interlayer spacing control [68], and the selected fabrication method [59] are the key factors influencing the fabrication of MXene-based membranes. Parameters including MXene concentration,

loading amount, drying conditions and polymer-to-MXene ratio influence the quality of dispersion, defect formation and uniformity of the transport channels. The morphology of the membrane, such as its lamellar ordering, porosity and surface roughness, is important in controlling water flux and solute rejection. The interlayer spacing is of particular significance for MXene membranes as it directly governs the transport through the nanochannels. Therefore, swelling is often avoided and selective pathways are preserved by intercalation, crosslinking and hybridisation. In addition, the manufacturing process of the membrane is also an important factor for the performance of the membranes, since vacuum filtration, coating, spray deposition, layer-by-layer assembly and polymer blending lead to different structural features, scalability and stability over time.

6- Critical discussion

MXenes are promising materials for water treatment applications owing to its hydrophilicity, tunable surface chemistry and layered nanochannel structure. However, their performance is highly dependent on the MXene type, the fabrication technique and the membrane architecture. $Ti_3C_2T_x$ is still the most investigated MXene and other MXenes are still unexplored. Vacuum filtration and other thin film formation methods work, but often suffer from restacking, swelling and scalability problems. Polymer-MXene composites enhance the mechanical stability and processability but most of the studies are conducted under ideal laboratory conditions, not under real wastewater. Future work should consider safer synthesis routes, long-term durability, antifouling performance, regeneration, and pilot-scale validation. Table 3 shows the critical evaluation of the application of $Ti_3C_2T_x$ and Mo-based MXenes in water treatment membranes

Table 3. Critical comparison of $Ti_3C_2T_x$ and Mo-based MXenes for membrane-based water treatment

Aspect	$Ti_3C_2T_x$	V_2C/Nb_2C / Mo-based MXenes	Main implication	Ref.
Material abundance	Most studied and best documented.	Much less studied.	$Ti_3C_2T_x$ has the strongest evidence base, but not necessarily the best performance for every target.	[50]
Surface chemistry	Strong hydrophilicity and common O/OH/F terminations.	Potentially different charge and reactivity profiles.	Composition strongly affects adsorption, ion sieving, and antifouling.	[73]
Fabrication compatibility	Easily integrated into vacuum-filtered films and polymer composites.	Feasible, but less frequently reported.	The best-studied material is also the easiest to benchmark.	[74]
Water flux	Often high in engineered composites. link.springer+1	Data are limited and less standardized.	Direct comparison is still weak.	[75]
Selectivity/rejection	Strong for dyes, ions, and some organics.	Promising but underexplored.	More composition-dependent benchmarking is needed.	[51]
Stability	Improved by polymer blending and crosslinking, but oxidation/swelling remain issues.	Similar or potentially different stability profiles, but data are sparse.	Long-term durability is still the main bottleneck.	[76]
Practical maturity	Highest among MXenes for water treatment.	Lower maturity.	$Ti_3C_2T_x$ is the benchmark, while others are promising future candidates.	[77]

7- Applications of MXene

7.1. Adsorption-based treatment

MXenes possess a large available surface area and reactive functional groups, which are effective for

pollutant adsorption [78]. Table 4 summarises that heavy metals, dyes, and some organic contaminants can be adsorbed onto the MXene surfaces by electrostatic attraction, coordination bonding, hydrogen bonding, and ion exchange. This renders them attractive for batch

adsorption and hybrid membrane-adsorber configurations [70].

The adsorption efficiency is greatly affected by pH, competing ions, contact time and natural organic matter [79]. MXenes show great adsorption capability in controlled laboratory conditions, but their performance can be hindered in real wastewater due to fouling or competition with background salts. From an economic point of view, the adsorbents should be used repeatedly, so the regeneration process is very important.

Consequently, supported membranes and composites are gradually replacing loose particles [80,81].

Functionalisation or chemical amalgamation can improve the adsorption of MXene [82]. For example, carbon-based or polymeric composites can improve the dispersion and the accessibility of binding sites [83]. This expands the range of contaminants that can be treated and enhances material recovery after use [84, 85]. One of the simplest and promising uses of MXenes is adsorption for water purification.

Table 4. Removal of different inorganic and organic pollutants by MXene-based membranes

Membrane composition	Pollutants	Removal Efficiency %	Operating Conditions	Ref.
$Ti_3C_2T_x$	Methylene Blue	> 99%	Dead-end filtration, 1 bar pressure, room temperature	[86]
	Oil	> 98.5%	Gravity-driven filtration, ambient pressure	[87]
Alkali- $Ti_3C_2T_x$ / PTFE	Pb^{2+} , Cu^{2+} , Cd^{2+}	95 – 99.2%	0.5 – 2 V external voltage assistance	[88]
$Ti_3C_2T_x$ –graphene oxide	Crystal violet	> 99.5%	Cross-flow filtration, 1 bar pressure	[89]
Nb_2CT_x	Dye	> 96%	Dead-end nanofiltration, 1 – 2 bar	[90]
V_2CT_x	oil–water separation	> 99%	Gravity-driven filtration, superhydrophilic wetting	[91]
$Ti_3C_2T_x$ -GO/PA	Li^+ , Na^+ , K^+ , Al^{3+}	High multi-valent ion rejection (~90% for Al^{3+})	2 – 5 bar pressure, selective size/charge sieving	[92]
$Ti_3C_2T_x$ /PAN	$NaCl$	65 – 85%	Nanofiltration mode, 3 bar driving pressure	[93]
$Ti_3C_2T_x$ /PVDF	$NaCl$ or $MgSO_4$	70 – 92%	4 – 6 bar pressure, 25°C feed water	[94]
$Ti_3C_2T_x$	E. coli / B. subtilis	> 99% microbial log reduction	Ambient contact filtration, membrane surface cell-killing	[95]

7.1.1. Dye removal

Industrial dyes are frequent contaminants in wastewater from the textile and printing industries [96]. MXene membranes are relevant to this problem because they combine adsorption and filtration in one platform [97]. Hydrophilic and negatively charged surfaces normally lead to rejection of cationic dyes [61, 98]. Dye molecules are larger than water molecules and can be separated by size exclusion [99]. Performance is determined by membrane thickness, pore structure, and interlayer spacing. Water flux reduction is significant in very tight nanochannels, while rejection may be lower in the case of nanochannels that are too wide. Thus, membrane design requires a delicate balance between permeability and selectivity. The composite techniques are effective for the preservation of equilibrium by strengthening the membrane and changing its inner architecture [16, 44, 100].

Many studies have demonstrated that MXene-based membranes outperform traditional polymer films when it comes to flow and rejection (especially when the membrane is properly engineered). The increased hydrophilicity can reduce the adsorption of hydrophobic pollutants, which can improve the long-term performance. The dye contamination is readily repeatable and detectable, thus commonly used as a testing platform for the study of MXene membranes [101].

7.1.2. Heavy metal removal

Lead, cadmium, chromium, copper and nickel are heavy metals with major environmental and health effects. These ions can be removed by electrostatic attraction, surface complexation and ion exchange with MXenes [102–104]. Such ions can be removed by electrostatic attraction, surface complexation, and ion exchange of MXenes [79]. This is especially the case for surfaces terminated by oxygen and hydroxyl groups that promote binding. MXenes have great potential to be used in membrane-based metal extraction and adsorption columns.

The efficiency of heavy-metal extraction is dependent on the solution chemistry and the oxidation state of MXene surface sites for metal cations [105, 106]. For instance, an increase in the salinity or competing ions can reduce the selectivity of adsorption. In practical applications, the material must resist fouling and retain its structural integrity under repeated use [81]. In such contexts, hybrid membranes and crosslinked films are usually more appropriate than pure MXene layers [107]. Another advantage is that MXene-based systems can be developed for selective separation rather than just bulk removal [108]. This shows that under some conditions they can discriminate between different kinds of ions based on their size, charge density or hydration energy. Such selectivity is especially useful in the treatment of

industrial wastewaters containing a mixture of metals. Therefore, MXenes are used as adsorbents and substrate materials for special ion-sieving membranes [109].

7.2. Desalination and ion sieving

The ion sieving and desalination potential of MXenes is mainly investigated [110]. Highly ordered nanochannels are formed by stacked MXene sheets, which enable precise control of water and ion transport [35]. MXenes have attracted researchers' attention as promising candidates for novel membranes in water desalination owing to their extremely elongated graphitic channels with nanoscale size [111]. MXenes-based membranes can be divided into two categories: freestanding MXenes and composite MXenes [112]. Compared to conventional desalination membranes, such as reverse osmosis and thin-film composite (TFC) membranes, MXenes-based membranes have enhanced performance [56]. MXenes enhance the fouling resistance, desalination efficiency, water permeability and other functions of membranes through their nanochannels, which allow molecular passage through "molecular-sieving". Thus, the membrane selectivity for the separation of specific molecules is improved.

One of the big advantages of MXenes in this respect is the interlayer spacing, which can be tuned. The dimensions of transport channels can be controlled by changes in hydration, intercalation and crosslinking. However, uncontrolled swelling in water is still a problem, as it can lead to an increase in channel space and a decrease in salt rejection. Thus, stabilisation techniques are essential in the design of desalination membranes [51].

Recent assessments suggest that MXene-based membranes hold great promise for advanced water purification and desalination, particularly when incorporated into composite or multilayer configurations [113]. Water transport is enabled by their high hydrophilicity, while the layered structure gives rise to size- and charge-selective channels [114]. However, long-term mechanical and chemical stability, which is an open question in the case of effective desalination, is required. This domain is one of the most dynamic and promising fields in MXene research and continues to be so [115].

7.3. Oil/water separation

Contaminants in industrial wastewater discharges must be eradicated before release, since they may present considerable long-term threats to human health and environmental safety [116]. Oily wastewater effluent discharge limitations vary by location, ranging from 5 to 100 mg/L [117]. MXene nanosheets were utilised to mitigate issues related to polarised layers, concentration polarisation, and specifically membrane fouling, hence improving the efficacy of separation procedures.

MXene-based membranes are being investigated for oil/water separation, especially in emulsified and surfactant-stabilised systems [118]. Such surfaces are

hydrophilic and oleophobic, and they can absorb water while rejecting oil droplets. MXene coatings and MXene-composite fabrics could be used for the clean-up of oily industrial wastes [119].

The underlying mechanism is based on surface wettability, size exclusion and structural selectivity. The effectiveness of the membrane depends on its ability to maintain anti-oil properties under conditions of pressure, salinity and frequent use. Durability and ease of handling are often improved by using composite structures. MXenes for oil/water separation are a promising application, which is practically important, but less developed than dye removal or desalination [61, 67].

7.4. Antibacterial and antifouling membranes

Viral or biological contamination is one of the most detrimental global challenges and a major contributor to membrane fouling, which has been associated with several waterborne infections [120]. Since biofouling influences membrane performance, it is important to study the antibacterial properties of membranes. Recently, many materials, such as MXenes, carbon-based compounds, metals and metal oxide nanoparticles, were investigated as bactericidal nanomaterials for evaluating their antibacterial effectiveness against Gram-negative and Gram-positive bacteria Fig. 5 [25].

Biofouling is a significant problem in water treatment, as microbial growth can reduce flux, block pores, and shorten the life of the membrane. MXenes are of particular interest because they can enhance antibacterial activity and are compatible with other antimicrobial agents. These are therefore advantageous for membranes for potable water and wastewater applications where resistance to microbial colonisation is required [121].

The particular antibacterial mechanism may be a function of MXene content, surface chemistry and composite properties. Researchers often combine MXenes with silver, polymers and other bioactive components to enhance their antibacterial performance. The hydrophilicity of MXene may reduce the adhesion of foulants, but it does not have direct bactericidal properties. This points to the capability of MXenes for both active and passive antifouling strategies [95, 122].

8- Composite and hybrid systems

The MXene-based composites and hybrid systems are gaining momentum as potential solutions for wastewater treatment. These systems utilise the unique properties of MXenes, such as their tunable porosity and hydrophilicity, to enhance the performance of membranes [16].

Membranes made of MXene composites are made by adding filler particles to polymer-based membranes, such as thin-film nanocomposites. MXene layers can enhance the membrane's hydrophilicity and selectivity and can be employed in water purification processes like UF (Ultrafiltration), NF (Nanofiltration), and RO (Reverse Osmosis) [114, 123].

The performance of MXene-based membranes can be improved by various modification strategies, such as regulation of MXene nanosheets, assembly processes of membranes, and surface modifications. These strategies are designed to control interlayer spacing, surface charge, wettability, and defects to improve the performance and efficiency of the membrane [57, 68].

MXene@ZIF-8 hybrid composites have exhibited strong adsorption efficiency for pollutants such as methyl orange and ciprofloxacin. The existence of non-covalent interactions between the pollutant molecules and the substrate [124] promotes effective removal of contaminations.

The progress in MXene-based systems has led to substantial development in sustainable and effective membrane-based separation methods for wastewater treatment.

One of the main trends in the MXene field is the transition from pure MXene films to hybrid materials. Polymer/MXene membranes are often used to overcome the problems of mechanical brittleness and swelling of pure layered films. These composites can be designed for simultaneous improvements in permeability, selectivity, flexibility and processability. They allow for mass production of membranes.

A common approach is to add supplementary nanofillers like graphene oxide, metal oxides, carbon nanotubes or other two-dimensional materials. The compounds can be used as interlayer pillars to increase structural stability or to impart catalytic and antibacterial properties. Composite design is more than a mechanical solution; it is a methodology for the design of multifunctional water-treatment materials. This region is expected to be important for practical applications of MXenes.

9- Limitations and challenges

MXenes have shown a lot of promise in laboratory testing, but there are many challenges to overcome before MXenes can be used widely in water treatment. Oxidation is a major challenge, especially for $Ti_3C_2T_x$, which is prone to degradation in time in air or aqueous conditions. Concerns of swelling and restacking can affect the structural integrity of nanochannels and the rejection efficiency. Pure MXene films have intrinsic mechanical brittleness, which limits their use as a stand-alone material [125].

MXene based materials are promising for water treatment, however some important challenges limit their practical application. Oxidation stability is one of the major concerns especially for $Ti_3C_2T_x$, which may degrade its properties during storage or operation in presence of oxygen, water and heat resulting in changes in conductivity, surface chemistry and overall membrane performance. Moreover, the restacking of MXene nanosheets into densely packed layers reduces the accessible surface area and transport pathways, which limits the water flux and reduces the number of active sites for adsorption or separation. Another major

limitation is swelling in aqueous media, which can enhance permeability temporarily through uncontrolled expansion of interlayer spacing but often at the expense of selectivity and structural stability. Another major issue with free-standing MXene films is their mechanical stability. They are brittle and can be prone to cracking under pressure, bending or multiple uses. They are frequently incorporated into polymer matrices or hybrid architectures for enhanced durability and handling. Besides these performance-related issues, toxicity and environmental safety of MXenes still require further studies, as recent studies have reported that their biological effects might depend on their composition, dose, surface chemistry and ageing behaviour. Finally, the disposal of used MXene-based materials is still an open issue, particularly when they are loaded with captured contaminants such as heavy metals or dyes. The development of regeneration, lifecycle management and environmentally friendly solutions at the end-of-life is required [77, 126].

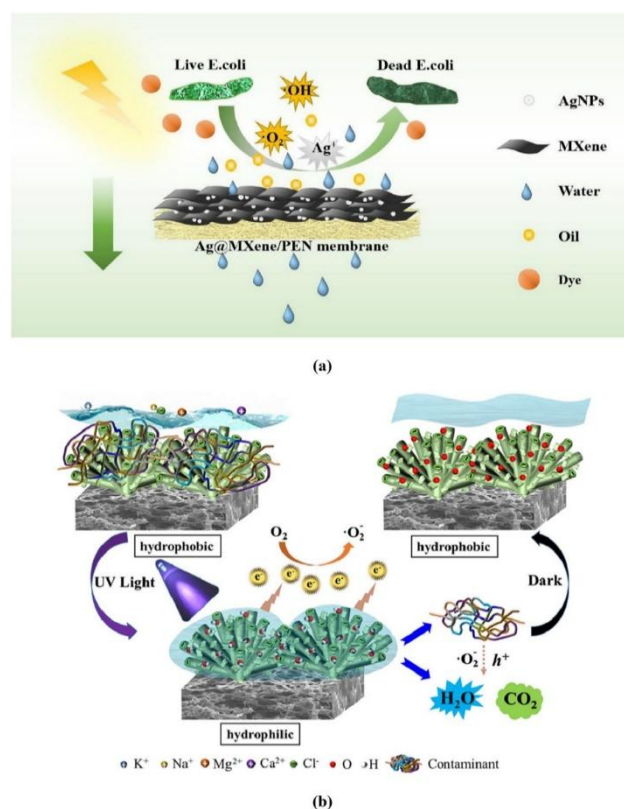


Fig. 5. (a) Antibacterial activity of MXene. (b) Antibacterial and fouling-resistant covalently cross-linked MXene-based membrane, reproduced from [25], licensed under CC BY 4.0

10- Future directions

Material design, process engineering, and scalability are key to the future application of MXenes for water treatment. Development of safer and more ecologically sustainable synthetic processes will be critical for mitigating the ecological challenges of fluoride-based etching. Long-term stability enhancement requires

improved oxidation control, cross-linking and composite formation. These advances will make the step from discovery in the laboratory to practical application.

Future research should focus on the development of membranes for specific applications such as desalination, metal recovery, pathogen removal, and industrial wastewater treatment. The use of real wastewater, pilot-scale systems and detailed fouling studies will be critical. At the same time, advanced hybrid systems combining adsorption, catalysis and membrane separation may have the highest practical value. MXenes are predicted to be a hot topic of interest in the field of membrane science and water treatment in the foreseeable future.

However, the industrial application of MXene membranes is still hindered by unsafe and non-scalable synthesis, oxidation and swelling, restacking, limited mechanical durability, and insufficient testing in real wastewater.

11- Conclusion

MXenes represent a highly promising category of two-dimensional materials for enhanced water treatment due to their extensive surface area, adjustable interlayer structure, hydrophilicity, and diverse surface chemistry. $Ti_3C_2T_x$ is the most extensively researched among them, although other MXene compositions remain comparatively underexplored. Although MXene-based systems exhibit promising performance in adsorption, desalination, oil-water separation, and antifouling membranes, their practical applications are hindered by sensitivity to oxidation, restacking, swelling, low mechanical strength, and difficulty in scaling up. Composite and hybrid architectures offer a more viable approach to achieving stable and processable membranes; nonetheless, their design must be meticulously optimised to maintain transport efficiency and structural integrity. Future advancements rely on enhanced synthesis methodologies, improved long-term stability, accurate assessment of wastewater, and scalable manufacturing techniques. A more rigorous and application-focused research emphasis is crucial for advancing MXene-based materials from laboratory demonstrations to practical water treatment systems.

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Conflict of Interest

The authors declare no conflict of interest.

References

- [1] A.K. Mohammed, S.M. Saadoon, Z.T. Abd Ali, I.M. Rashid, N. Hussin AL Sbani, "Removal of amoxicillin from contaminated water using modified bentonite as a reactive material," *Heliyon*. 10, 2024. <https://doi.org/10.1016/j.heliyon.2024.e24916>
- [2] M.M.H. Hammad, K.W. Hameed, H.A. Sabti, "Reducing the Pollutants from Municipal Wastewater by *Chlorella Vulgaris* Microalgae," *Al-Khwarizmi Engineering Journal*. 15, 2019, 97–108. <https://doi.org/10.22153/kej.2019.09.001>
- [3] R. Sangamnere, T. Misra, H. Bherwani, A. Kapley, R. Kumar, "A critical review of conventional and emerging wastewater treatment technologies," *Sustainable Water Resources Management*. 9, 2023, 58. <https://doi.org/10.1007/s40899-023-00829-y>
- [4] K. Ji, Z. Liang, P. Wang, Z. Li, Q. Ma, X. Su, "Mxene-based capacitive enzyme-free biofuel cell Self-Powered sensor for lead ion detection in human plasma," *Chemical Engineering Journal*. 495, 2024, 153598. <https://doi.org/10.1016/J.CEJ.2024.153598>
- [5] D. Dhamodharan, M.A. Al-Harhi, B. Ramya, A. Bafaqeer, F. Alam, "MXenes: A promising material with multifunctional applications," *Journal of Environmental Chemical Engineering*. 12, 2024, 112316. <https://doi.org/10.1016/J.JECE.2024.112316>
- [6] V. Hada, D. Malvi, M. Mili, M.M. Khan, G. Chaturvedi, S.A.R. Hashmi, A.K. Srivastava, S. Verma, "MXenes: promising 2D materials for wound dressing applications - a perspective review," *Materials Advances*. 3, 2022, 7445–7462. <https://doi.org/10.1039/d1ma01199e>
- [7] S. Kumar, N. Kumari, Y. Seo, "MXenes: Versatile 2D materials with tailored surface chemistry and diverse applications," *Journal of Energy Chemistry*. 90, 2024, 253–293. <https://doi.org/10.1016/J.JECHEM.2023.11.031>
- [8] T.K. Paul, M.A. Khaleque, M.R. Ali, M. Aly Saad Aly, M.S. Bacchu, S. Rahman, M.Z.H. Khan, "MXenes from MAX phases: synthesis, hybridization, and advances in supercapacitor applications," *RSC Advances*. 15, 2025, 8948–8976. <https://doi.org/10.1039/d5ra00271k>
- [9] L. Liu, H. Zschiesche, M. Antonietti, M. Gibilaro, P. Chamelot, L. Massot, P. Rozier, P.L. Taberna, P. Simon, "In Situ Synthesis of MXene with Tunable Morphology by Electrochemical Etching of MAX Phase Prepared in Molten Salt," *Advanced Energy Materials*. 13, 2023. <https://doi.org/10.1002/aenm.202203805>
- [10] Y. Liu, Y. Wu, X. Wang, "Thermal transports in the MXenes family: Opportunities and challenges," *Nano Research*. 17, 2024, 7700–7716. <https://doi.org/10.1007/s12274-024-6763-6>

- [11] Z. Said, M.A. Sohail, H. Tabassum, I.H. Sajid, H.M. Ali, F. Jamil, Y.K. Mishra, A.K. Pandey, "MXenes at the forefront: advances in energy storage and nanofluidic applications," *Springer International Publishing*, 2025. <https://doi.org/10.1007/s42114-025-01404-z>
- [12] R. Ghanbari, E. Nazarzadeh Zare, "Engineered MXene-polymer composites for water remediation: Promises, challenges and future perspective," *Coordination Chemistry Reviews*. 518, 2024, 216089. <https://doi.org/10.1016/J.CCR.2024.216089>
- [13] Y. Sun, J. Lu, S. Li, C. Dai, D. Zou, W. Jing, "MXene-based membranes in water treatment: Current status and future prospects," *Separation and Purification Technology*. 331, 2024, 125640. <https://doi.org/10.1016/J.SEPUR.2023.125640>
- [14] T. Zhang, Z. Huang, D. Cui, Z. Luo, X. Zhuo, Y. Liu, X. Wang, K. Wu, J. Zhang, G. Li, S. Sun, H. Zuilhof, H. Lu, "2D lamellar membranes with defect-engineered hierarchical channels impart ultrafast and selective dye/salt separation," *Journal of Membrane Science*. 735, 2025, 124602. <https://doi.org/10.1016/j.memsci.2025.124602>
- [15] C. Qiao, H. Wu, X. Xu, Z. Guan, W. Ou-Yang, "Electrical Conductivity Enhancement and Electronic Applications of 2D Ti₃C₂T_x MXene Materials," *Advanced Materials Interfaces*. 8, 2021, 2100903. <https://doi.org/10.1002/admi.202100903>
- [16] Z. Hu, Y. Li, A. Li, H.H. Wang, X.F. Wang, "Dye-sensitized solar cells based on highly catalytic CNTs/Ti₃C₂T_x MXenes composite counter electrode," *RSC Advances*. 13, 2023, 34808–34816. <https://doi.org/10.1039/d3ra06814e>
- [17] T. Jabeen, M. Rashid, S. Haider, M. Azam, "MXenes-based nanofiltration membranes for improved desalination performance: Recent advances and future prospects," *Journal of Hazardous Materials Advances*. 19, 2025, 100814. <https://doi.org/10.1016/J.HAZADV.2025.100814>
- [18] S. Jayakumar, P.C. Santhosh, S. Ramakrishna, A. V. Radhamani, "2D (Ti₃C₂T_x) MXene: A comprehensive review of advancements in synthesis protocols, applications in supercapacitors, sustainability targets and future prospects," *Journal of Energy Storage*. 97, 2024, 112741. <https://doi.org/10.1016/J.EST.2024.112741>
- [19] Z.W.S. Yuanjian Li, Xiang Feng, Wei Ying Lieu, Lin Fu, Chang Zhang, Tanmay Ghosh, Anupma Thakur, Brian C. Wyatt, Babak Anasori, Wei Liu, Qianfan Zhang, Jun Lu, "MXene-Based Anode-Free Magnesium Metal Battery," *Advanced Functional Materials*. 33, 2023. <https://doi.org/10.1002/adfm.202303067>
- [20] B. Pant, M. Park, A.A. Kim, "MXene-Embedded Electrospun Polymeric Nanofibers for Biomedical Applications: Recent Advances," *Micromachines*. 14, 2023, 1–19. <https://doi.org/10.3390/mi14071477>
- [21] N. Goel, A. Kushwaha, M. Kumar, "Two-dimensional MXenes: recent emerging applications," *RSC Advances*. 12, 2022, 25172–25193. <https://doi.org/10.1039/d2ra04354h>
- [22] A. Koyappayil, S.G. Chavan, Y.-G. Roh, M.-H. Lee, "Advances of MXenes; Perspectives on Biomedical Research," *Biosensors*. 12, 2022. <https://doi.org/10.3390/bios12070454>
- [23] K.A. Papadopoulou, A. Chronos, S.-R.G. Christopoulos, "Latest advances and comparative analysis of MXenes as anode and cathode electrodes in secondary batteries," *Journal of Applied Physics*. 133, 2023, 30901. <https://doi.org/10.1063/5.0136840>
- [24] F. Dixit, K. Zimmermann, R. Dutta, N.J. Prakash, B. Barbeau, M. Mohseni, B. Kandasubramanian, "Application of MXenes for water treatment and energy-efficient desalination: A review," *Journal of Hazardous Materials*. 423, 2022, 127050. <https://doi.org/10.1016/j.jhazmat.2021.127050>
- [25] H. Meskher, A.K. Thakur, S.K. Hazra, M.S. Ahamed, A.M. Saleque, Q.F. Alsahy, M.W. Shahzad, M.N. Al Subri Ivan, S. Saha, I. Lynch, "Recent advances in applications of MXenes for desalination, water purification and as an antibacterial: a review," *Environmental Science: Nano*. 12, 2025, 1012–1036. <https://doi.org/10.1039/d4en00427b>
- [26] Z.E. AlHadithy, A.A. AbdulRazak, A.M.H.A. Al-Ghaban, Q.F. Alsahy, H. Meskher, R.A. Al-Juboori, K.K. Katibi, D.U. Lawal, "Advancements in Water Treatment with MXene-Enhanced Membranes: A Review of Current Progress and Future Directions," *Springer International Publishing*, 2024. <https://doi.org/10.1007/s11270-024-07628-x>
- [27] A.M. Amani, M. Abbasi, A. Najdian, F. Mohamadpour, S.R. Kasaei, H. Kamyab, S. Chelliapan, M. Shafiee, L. Tayebi, A. Vaez, A. Najafian, E. Vafa, S. Mosleh-Shirazi, "MXene-based materials for enhanced water quality: Advances in remediation strategies," *Ecotoxicology and Environmental Safety*. 291, 2025, 117817. <https://doi.org/10.1016/j.ecoenv.2025.117817>
- [28] C. Bhuyan, P. Bora, S. Hazarika, "MXenes in membrane engineering: Unlocking the new potential for water treatment," *Water Supply*. 25, 2025, 1209–1234. <https://doi.org/10.2166/ws.2025.072>
- [29] E.S. Sowbakkiyavathi, P. Dhandapani, M. Neelakandan, G. Murali, I. In, S.J. Lee, A. Subramania, "A comprehensive review on MXenes: synthesis, stability, properties and their functionalization for M-ion batteries and supercapacitors," *Nanoscale*. 18, 2026, 640–693. <https://doi.org/10.1039/d5nr03760c>
- [30] M.I.H. Protayi, A. Bin Rashid, "A comprehensive overview of recent progress in MXene-based polymer composites: Their fabrication processes, advanced applications, and prospects," *Heliyon*. 10, 2024, e37030. <https://doi.org/10.1016/j.heliyon.2024.e37030>

- [31] Y. Jiang, "Applications and perspectives of Ti₃C₂T_x MXene in electrochemical energy storage systems," *International Journal of Electrochemical Science*. 20, 2025, 100948. <https://doi.org/10.1016/j.ijoes.2025.100948>
- [32] Soni, M. Ahuja, P.K. Jagtap, V. Chauhan, S.K. Raj, P.P. Sharma, "The Advent of MXene-Based Synthetics and Modification Approaches for Advanced Applications in Wastewater Treatment," *Membranes*. 15, 2025. <https://doi.org/10.3390/membranes15120364>
- [33] A. Bathla, "Implementation of MXenes for Water Treatment, in: O.P. Pandey, P. Sharma (Eds.), MXenes: Emerging 2D Materials, Springer Nature Singapore," *Singapore*, 2024, pp. 109–119. https://doi.org/10.1007/978-981-97-4064-2_6
- [34] N.-S. Mussa, K. Askaruly, K. Bexetova, S. Azat, K. Toshtay, "Recent advancements in MXene-based catalysts: Synthesis, characterization, and applications in sustainable energy production," *Carbon Trends*. 20, 2025, 100551. <https://doi.org/10.1016/j.cartre.2025.100551>
- [35] Y. Sun, J. Lu, S. Li, C. Dai, D. Zou, W. Jing, "MXene-based membranes in water treatment: Current status and future prospects," *Separation and Purification Technology*. 331, 2024, 125640. <https://doi.org/10.1016/j.seppur.2023.125640>
- [36] P.O.Å. Persson, J. Rosen, "Current state of the art on tailoring the MXene composition, structure, and surface chemistry," *Current Opinion in Solid State and Materials Science*. 23, 2019, 100774. <https://doi.org/10.1016/j.cossms.2019.100774>
- [37] X. Gao, Z. Li, J. Xue, Y. Qian, L. Zhang, J. Caro, H. Wang, "Titanium carbide Ti₃C₂T_x (MXene) enhanced PAN nanofiber membrane for air purification," *Journal of Membrane Science*. 586, 2019, 162–169. <https://doi.org/10.1016/j.memsci.2019.05.058>
- [38] N. Bisht, S. Jaiswal, J. Vishwakarma, S.K. Gupta, R.J. Yeo, S.K.R.S. Sankaranarayanan, C. Dhand, N. Dwivedi, "Mxene enhanced Shape Memory Polymer Composites: The rise of MXenes as fillers for stimuli-responsive materials," *Chemical Engineering Journal*. 498, 2024. <https://doi.org/10.1016/j.cej.2024.155154>
- [39] Z. Lu, Y. Wei, J. Deng, L. Ding, Z.-K. Li, H. Wang, "Self-Crosslinked MXene (Ti₃C₂T_x) Membranes with Good Antiswelling Property for Monovalent Metal Ion Exclusion," *ACS Nano*. 13, 2019, 10535–10544. <https://doi.org/10.1021/acsnano.9b04612>
- [40] Hou, H. Zhou, K. Zhu, C. Xie, S. Liu, Y. He, X. Zhu, M. Wu, T. Huang, "Superwetting Ti₃C₂T_x MXene membranes intercalated with sodium alginate for oil/water separation," *Carbohydrate Polymer Technologies and Applications*. 5, 2023, 100278. <https://doi.org/10.1016/J.CARPTA.2022.100278>
- [41] Qian, C. Yuan, X. Wang, H. Zhang, L. Du, G. Wei, S. Chen, "Conductive MXene ultrafiltration membrane for improved antifouling ability and water quality under electrochemical assistance," *RSC Advances*. 13, 2023, 15872–15880. <https://doi.org/10.1039/d3ra01116j>
- [42] S. Jin, Y. Guo, F. Wang, A. Zhou, "The synthesis of MXenes," *MRS Bulletin*. 48, 2023, 245–252. <https://doi.org/10.1557/s43577-023-00491-x>
- [43] J. Michael, Z. Qifeng, W. Danling, "Titanium carbide MXene: Synthesis, electrical and optical properties and their applications in sensors and energy storage devices," *Nanomaterials and Nanotechnology*. 9, 2019, 1–9. <https://doi.org/10.1177/1847980418824470>
- [44] H. Kiran, A.B.M. Ali, R. Ashraf, M.B. Tahir, M. Nasir, N. Hasnain, M.F. Ullah, D. Abduvalieva, N. Batool, "Comparative Study of Ti₃C₂T_x MXene Synthesized via HF and LiF-HCl Etching: Structural, Optical, and Photocatalytic Insights," *Journal of Inorganic and Organometallic Polymers and Materials*. 35, 2025, 9565–9578. <https://doi.org/10.1007/s10904-025-03832-2>
- [45] S. Nouseen, M. Pumera, "Electrochemical etching of MXenes: mechanism, challenges and future outlooks," *Journal of Materials Chemistry A*. 13, 2025, 34055–34084. <https://doi.org/10.1039/d5ta04176g>
- [46] F. Wang, S. Wang, F. Tian, F. Wang, X. Xia, Q. Zhang, Z. Pang, X. Yu, G. Li, H.-Y. Hsu, S. Hu, L. Ji, Q. Xu, Y. Zhao, X. Zou, X. Lu, "Advances in molten-salt-assisted synthesis of 2D MXenes and their applications in electrochemical energy storage and conversion," *Chemical Engineering Journal*. 470, 2023, 144185. <https://doi.org/10.1016/j.cej.2023.144185>
- [47] Mahabari, R.D. Mohili, M. Patel, A.H. Jadhav, K. Lee, N.K. Chaudhari, "HF-free microwave-assisted synthesis of MXene as an electrocatalyst for hydrogen evolution in alkaline media," *Nanoscale Advances*. 6, 2024, 5388–5397. <https://doi.org/10.1039/D4NA00250D>
- [48] W. Huang, J. Wang, W. Lai, M. Guo, "MXene Surface Architectonics: Bridging Molecular Design to Multifunctional Applications," *Molecules*. 30, 2025. <https://doi.org/10.3390/molecules30091929>
- [49] T. Amrillah, C.A.C. Abdullah, A. Hermawan, F.N.I. Sari, V.N. Alvani, "Towards Greener and More Sustainable Synthesis of MXenes: A Review," *Nanomaterials*. 12, 2022. <https://doi.org/10.3390/nano12234280>
- [50] A. Keneshbekova, G. Smagulova, B. Kaidar, A. Imash, A. Ilyanov, R. Kazhdanbekov, E. Yensep, A. Lesbayev, "MXene/Carbon Nanocomposites for Water Treatment," *Membranes*. 14, 2024. <https://doi.org/10.3390/membranes14090184>

- [51] H. Jain, L. Singh, A. Shrivastava, M. Paul, H. Horo, H.S. Samuel, "MXene-enhanced polymer nanocomposite membranes for next-generation wastewater treatment and desalination," *Next Materials*. 10, 2026, 101574. <https://doi.org/10.1016/j.nxmate.2025.101574>
- [52] K. Maleski, M. Alhabeib, "Top-Down MXene Synthesis (Selective Etching), in: B. Anasori, Y. Gogotsi (Eds.), 2D Metal Carbides and Nitrides (MXenes): Structure, Properties and Applications," *Springer International Publishing, Cham*, 2019, pp. 69–87. <https://doi.org/10.1007/978-3-030-19026-2>
- [53] Ahmadi Bonakdar, D. Rodrigue, M.A. Bonakdar, D. Rodrigue, "Electrospinning: Processes, Structures, and Materials," *Macromol.* 4, 2024, 58–103. <https://doi.org/10.3390/macromol4010004>
- [54] Al-Furaiji, M. Kadhom, K. Kalash, B. Waisi, N. Albayati, "Preparation of thin-film composite membranes supported with electrospun nanofibers for desalination by forward osmosis," *Drinking Water Engineering and Science*. 13, 2020, 51–57. <https://doi.org/10.5194/dwes-13-51-2020>
- [55] M. Mozafari, M. Soroush, "Surface functionalization of MXenes," *Materials Advances*. 2, 2021, 7277–7307. <https://doi.org/10.1039/d1ma00625h>
- [56] I.A. Khan, L.T. Yogarathinam, H. Younas, A. Mushtaq, I.H. Aljundi, N. Baig, "The rise of MXenes: Synthesis, properties, and fabrication of advanced membranes," *Desalination*. 620, 2026, 119598. <https://doi.org/10.1016/j.desal.2025.119598>
- [57] Y. Han, J. Hu, X. Liu, F. Liu, "Progress in Surface and Interface Modification Strategies of MXene Materials for Energy Storage Applications," *Materials*. 18, 2025, 1–33. <https://doi.org/10.3390/ma18153576>
- [58] R. Patel, K. Mahabari, K. Patel, M. Patel, A.H. Jadhav, S. Dharaskar, R.R. Karri, N. Chaudhari, "MoS₂/MXene composite material for high-performance removal of Rhodamine B dye: synthesis and application," *Scientific Reports*. 15, 2025, 42099. <https://doi.org/10.1038/s41598-025-26240-5>
- [59] A. Sreekumar, A.R. Nair, A. Raman, A. Sivan, M. Pandey, K. Deshmukh, S. Appukuttan, "MXene-Reinforced Polymer Composite Membranes for Water Desalination and Wastewater Treatment," *Wiley Semiconductors*, 2024, 459-500. <https://doi.org/10.1002/9781119901280.ch14>
- [60] Z. Huang, D. Ling Zhao, L. Shen, H. Lin, C. Chen, Y. Xu, B. Li, J. Teng, L. Han, T.-S. Chung, "MXenes for membrane separation: from fabrication strategies to advanced applications," *Science Bulletin*. 69, 2024, 125–140. <https://doi.org/10.1016/j.scib.2023.11.008>
- [61] R. Imsong, D. Dhar Purkayastha, "Dual-functional superhydrophilic/underwater superoleophobic 2D Ti₃C₂TX MXene-PAN membrane for efficient oil-water separation and adsorption of organic dyes in wastewater," *Separation and Purification Technology*. 306, 2023, 122636. <https://doi.org/10.1016/j.seppur.2022.122636>
- [62] Riaz, Z. Hussain, H. Pan, Z. Lin, "Engineering MXenes for advanced organic coatings: Synthesis, surface functionalization, corrosion protection performance, and future perspectives," *Progress in Organic Coatings*. 217, 2026, 110219. <https://doi.org/10.1016/j.porgcoat.2026.110219>
- [63] M. Danish, M. Tayyab Raza, M. Iftikhar, I. Hassan, S.A. Husnain, M. Muneeb ul Hassan, M. Mumtaz, A.M. Khan, H. Noor, F. Shahzad, "Spray deposited Graphene@MXene on a flexible polymer substrate for Terahertz Shielding," *Materials Chemistry and Physics*. 311, 2024, 128573. <https://doi.org/10.1016/j.matchemphys.2023.128573>
- [64] S. Hu, P. Han, S. Zhang, J. Wang, C. Meng, G. Wei, Z. Gu, "Layer-by-layer assembling MXene /poly(3-glycidyloxypropyldimethoxymethylsilane) hierarchical structure onto carbon fibers for high-performance carbon fiber/epoxy composites," *Composites Science and Technology*. 241, 2023, 110149. <https://doi.org/10.1016/j.compscitech.2023.110149>
- [65] S. Chen, Y. Du, X. Zhang, Y. Xie, Z. Shi, H. Ji, W. Zhao, C. Zhao, "One-step electrospinning of negatively-charged polyethersulfone nanofibrous membranes for selective removal of cationic dyes," *Journal of the Taiwan Institute of Chemical Engineers*. 82, 2018, 179–188. <https://doi.org/10.1016/J.JTICE.2017.11.018>
- [66] X. Gao, Z.K. Li, J. Xue, Y. Qian, L.Z. Zhang, J. Caro, H. Wang, "Titanium carbide Ti₃C₂TX (MXene) enhanced PAN nanofiber membrane for air purification," *Journal of Membrane Science*. 586, 2019, 162–169. <https://doi.org/10.1016/j.memsci.2019.05.058>
- [67] L. Zhang, Y. Liu, G. Zeng, Z. Yang, Q. Lin, Y. Wang, X. Wang, S. Pu, "Two-dimensional Na-Bentonite@MXene composite membrane with switchable wettability for selective oil/water separation," *Separation and Purification Technology*. 306, 2023, 122677. <https://doi.org/10.1016/J.SEPPUR.2022.122677>
- [68] Y. Du, J. Yu, B. Chen, X. Zhu, "Recent progresses in the modification strategies of MXene-based membranes for water and wastewater treatments," *Environmental Science: Nano*. 12, 2025, 150–188. <https://doi.org/10.1039/D4EN00712c>
- [69] H. Zhang, M. Yong, T. Hu, Y. Kang, Z. Wang, Z. Xu, X. Li, X. Sun, L. Guo, F. Sheng, X. Zeng, Z. Li, X. Li, H. Wang, T. Xu, X. Zhang, "Multifunctional intercalants create stable subnanochannels in MoS₂ membranes for wastewater treatment," *Nature Communications*. 16, 2025, 8353. <https://doi.org/10.1038/s41467-025-58409-x>
- [70] M.H. Munshi, S.A. Hussain, M. Akter, M.A. Hakim, M.A. Hasan, M.J. Kabir, M. Kamruzzaman, T. Islam, "MXene-Based Polymer Composites for Advanced Filtration: Properties, Fabrication, and Future Directions," *ChemistrySelect*. 11, 2026. <https://doi.org/10.1002/slct.202506512>

- [71] A.A. Bankole, R.S.B. Zakari, K. Ahmad, M. Abi Jaoude, "Challenges and Opportunities Associated with the Development of Functionalized Electrospun Fibers, in: S. Garg, A. Chandra, S. Sagadevan (Eds.), Emerging Sustainable Nanomaterials for Biomedical Applications," *Springer Nature Switzerland, Cham*, 2024, pp. 301–331. https://doi.org/10.1007/978-3-031-63961-6_12
- [72] H.S. Al-Okaidy, B.I. Waisi, "The Effect of Electrospinning Parameters on Morphological and Mechanical Properties of PAN-based Nanofibers Membrane," *Baghdad Science Journal*. 20, 2023, 1433–1441. <https://doi.org/10.21123/bsj.2023.7309>
- [73] P.O.Å. Persson, MXene Surface Chemistry, in: B. Anasori, Y. Gogotsi (Eds.), "2D Metal Carbides and Nitrides (MXenes): Structure, Properties and Applications," *Springer International Publishing, Cham*, 2019, pp. 125–136. https://doi.org/10.1007/978-3-030-19026-2_8
- [74] Q. Zhao, S. Huang, X. Han, J. Chen, X. Ma, "Highly active and controllable MOF-derived carbon nanosheets supported iron catalysts for Fischer–Tropsch synthesis," *Carbon*. 173, 2021. <https://doi.org/10.1016/j.carbon.2020.11.019>
- [75] Q. Feng, Y. Zhan, W. Yang, H. Dong, A. Sun, L. Li, X. Chen, Y. Chen, "Ultra-high flux and synergistically enhanced anti-fouling Ag@MXene lamellar membrane for the fast purification of oily wastewater through nano-intercalation, photocatalytic self-cleaning and antibacterial effect," *Separation and Purification Technology*. 298, 2022, 121635. <https://doi.org/10.1016/j.seppur.2022.121635>
- [76] M. Wu, B. Wang, Q. Hu, L. Wang, A. Zhou, "The Synthesis Process and Thermal Stability of V2C MXene," *Materials*. 11, 2018. <https://doi.org/10.3390/ma11112112>
- [77] O.J. Kewate, I. Hussain, N. Tyagi, S. Saxena, K. Zhang, E.G. Rajamansingh, N. Chinnappan, H. Joshi, S. Punniyakoti, "Nb-based MXenes: Structures, properties, synthesis, and application towards supercapacitors," *Journal of Energy Storage*. 94, 2024, 112445. <https://doi.org/10.1016/j.est.2024.112445>
- [78] S. Kumar, M. Taunk, "2D MXenes for flexible device applications," *Materials Today Physics*. 46, 2024, 101483. <https://doi.org/10.1016/j.mtphys.2024.101483>
- [79] M. Ilyas, M. Younas, M.U.H. Shah, W.U. Rehman, A.U. Rehman, Z.H. Yuan, Y.M. Zheng, M. Sheikh, M. Rezakazemi, "MXene-based 2D Ti3C2Tx nanosheets for highly efficient cadmium (Cd²⁺) adsorption," *Journal of Water Process Engineering*. 55, 2023, 104131. <https://doi.org/10.1016/j.jwpe.2023.104131>
- [80] M. Kaur, U. Krishnan, P. Sharma, K. Kaur, R.K. Sharma, A. Kumar, "MXenes in action: Adsorption strategies for environmental contaminants," *Journal of Alloys and Compounds*. 1034, 2025, 181350. <https://doi.org/10.1016/j.jallcom.2025.181350>
- [81] Z. Othman, H.R. Mackey, K.A. Mahmoud, "A critical overview of MXenes adsorption behavior toward heavy metals," *Chemosphere*. 295, 2022, 133849. <https://doi.org/10.1016/j.chemosphere.2022.133849>
- [82] Karthikeyan, S.S. Elanchezhian, J. Preethi, K. Talukdar, S. Meenakshi, C.M. Park, "Two-dimensional (2D) Ti3C2Tx MXene nanosheets with superior adsorption behavior for phosphate and nitrate ions from the aqueous environment," *Ceramics International*. 47, 2021, 732–739. <https://doi.org/10.1016/j.ceramint.2020.08.183>
- [83] S. Sharma, M. Mishra, H. Sharma, A.S. Chauhan, B. Das, D. Rajput, V. Pandey, R.B. Tripathi, R. Sharma, "Polymeric nanocomposites: Recent advances, challenges, techniques, and biomedical applications," *Annales Pharmaceutiques Françaises*. 83, 2025, 1019–1039. <https://doi.org/10.1016/j.pharma.2025.06.003>
- [84] B.M. Jun, M. Jang, C.M. Park, J. Han, Y. Yoon, "Selective adsorption of Cs⁺ by MXene (Ti3C2Tx) from model low-level radioactive wastewater," *Nuclear Engineering and Technology*. 52, 2020, 1201–1207. <https://doi.org/10.1016/j.net.2019.11.020>
- [85] Z.H. Zhang, J.Y. Xu, X.L. Yang, "MXene/sodium alginate gel beads for adsorption of methylene blue," *Materials Chemistry and Physics*. 260, 2021, 124123. <https://doi.org/10.1016/j.matchemphys.2020.12.4123>
- [86] Z. Wei, Z. Peigen, T. Wubian, Q. Xia, Z. Yamei, S. ZhengMing, "Alkali treated Ti3C2Tx MXenes and their dye adsorption performance," *Materials Chemistry and Physics*. 206, 2018, 270–276. <https://doi.org/10.1016/j.matchemphys.2017.12.034>
- [87] Sun, F. Zheng, M. Zhu, Z. Song, K. Wang, M. Zhong, D. Wu, R.B. Little, Z. Xu, H. Zhu, "Selective Trans-Membrane Transport of Alkali and Alkaline Earth Cations through Graphene Oxide Membranes Based on Cation– π Interactions," *ACS Nano*. 8, 2014, 850–859. <https://doi.org/10.1021/nm4055682>
- [88] S. Wang, F. Wang, Y. Jin, X. Meng, B. Meng, N. Yang, J. Sunarso, S. Liu, "Removal of heavy metal cations and co-existing anions in simulated wastewater by two separated hydroxylated MXene membranes under an external voltage," *Journal of Membrane Science*. 638, 2021, 119697. <https://doi.org/10.1016/j.memsci.2021.119697>
- [89] Y. Sun, Z. Xu, Y. Zhuang, G. Liu, W. Jin, G. Liu, W. Jing, "Tunable dextran retention of MXene-TiO2 mesoporous membranes by adjusting the 2D MXene content," *2D Materials*. 5, 2018, 45003. <https://doi.org/10.1088/2053-1583/aad01e>
- [90] Y. Yan, H. Han, Y. Dai, H. Zhu, W. Liu, X. Tang, W. Gan, H. Li, "Nb2CTx MXene Nanosheets for Dye Adsorption," *ACS Applied Nano Materials*. 4, 2021, 11763–11769. <https://doi.org/10.1021/acsanm.1c02339>

- [91] Y. Abraham Swu, D. Dhar Purkayastha, "V2CTx MXene for sustainable wastewater treatment: Inspired by the exoskeletons of nature's microscopic architects," *Applied Surface Science*. 680, 2025, 161365.
<https://doi.org/10.1016/j.apsusc.2024.161365>
- [92] Z. Yin, Z. Lu, Y. Xu, Y. Zhang, L. He, P. Li, L. Xiong, L. Ding, Y. Wei, H. Wang, "Supported MXene/GO Composite Membranes with Suppressed Swelling for Metal Ion Sieving," *Membranes*. 11, 2021. <https://doi.org/10.3390/membranes11080621>
- [93] G. Liu, J. Shen, Q. Liu, G. Liu, J. Xiong, J. Yang, W. Jin, "Ultrathin two-dimensional MXene membrane for pervaporation desalination," *Journal of Membrane Science*. 548, 2018, 548–558. <https://doi.org/10.1016/j.memsci.2017.11.065>
- [94] C.E. Ren, M. Alhabeb, B.W. Byles, M.-Q. Zhao, B. Anasori, E. Pomerantseva, K.A. Mahmoud, Y. Gogotsi, "Voltage-Gated Ions Sieving through 2D MXene Ti3C2Tx Membranes," *ACS Applied Nano Materials*. 1, 2018, 3644–3652. <https://doi.org/10.1021/acsanm.8b00762>
- [95] Rasool, K.A. Mahmoud, D.J. Johnson, M. Helal, G.R. Berdiyrov, Y. Gogotsi, "Efficient Antibacterial Membrane based on Two-Dimensional Ti3C2Tx (MXene) Nanosheets," *Scientific Reports*. 7, 2017, 1598. <https://doi.org/10.1038/s41598-017-01714-3>
- [96] N.M. Jabbar, S.D. Salman, I.M. Rashid, Y.S. Mahdi, "Removal of an anionic Eosin dye from aqueous solution using modified activated carbon prepared from date palm fronds," *Chemical Data Collections*. 42, 2022, 100965. <https://doi.org/10.1016/j.cdc.2022.100965>
- [97] B.-M. Jun, J. Heo, N. Taheri-Qazvini, C.M. Park, Y. Yoon, "Adsorption of selected dyes on Ti3C2Tx MXene and Al-based metal-organic framework," *Ceramics International*. 46, 2020, 2960–2968. <https://doi.org/10.1016/j.ceramint.2019.09.293>
- [98] K.E. Talib, S.D. Salman, "Removal of Malachite Green from Aqueous Solution using Ficus Benjamina Activated Carbon-Nonmetal Oxide synthesized by pyro Carbonic Acid Microwave," *Al-Khwarizmi Engineering Journal*. 19, 2023, 26–38. <https://doi.org/10.22153/kej.2023.03.002>
- [99] Lin, W. Ye, M. Xie, D.H. Seo, J. Luo, Y. Wan, B. Van der Bruggen, "Environmental impacts and remediation of dye-containing wastewater," *Nature Reviews Earth & Environment*. 4, 2023, 785–803. <https://doi.org/10.1038/s43017-023-00489-8>
- [100] C. Peng, P. Wei, X. Chen, Y. Zhang, F. Zhu, Y. Cao, H. Wang, H. Yu, F. Peng, "A hydrothermal etching route to synthesis of 2D MXene (Ti3C2, Nb2C): Enhanced exfoliation and improved adsorption performance," *Ceramics International*. 44, 2018, 18886–18893. <https://doi.org/10.1016/J.CERAMINT.2018.07.124>
- [101] V.N. Sahith, A. Kumar J, V.S. Sruthi, T. Krithiga, S. Sundararaman, P. Deivasigamani, D. Venkatesan, M. Anish, A. Hosseini-Bandegharai, "Mxenes-based sorbents: Pioneering nanomaterials and its mechanism for advanced dye remediation in aqueous environments," *Next Materials*. 9, 2025, 101029. <https://doi.org/10.1016/j.nxmte.2025.101029>
- [102] Q.A. Hind S. AbdulhayHameed, "Bioremoval of Chromium by Local Isolates of *Pseudomonas aeruginosa* in Respect to its Genotype," *Iraqi Journal of Sciences*. 57, 2016.
- [103] T. Liu, J.H. Hou, J.B. Wang, W. Wang, X.Y. Wang, J.L. Wu, "Biosorption of heavy metals from aqueous solution by the novel biosorbent *Pectobacterium* sp. ND2," *Environmental Progress and Sustainable Energy*. 37, 2018, 968–974. <https://doi.org/10.1002/ep.12757>
- [104] W. Khalid, S.D. Salman, "Adsorption of Heavy Metals from Aqueous Solution onto Sawdust Activated Carbon," *Al-Khwarizmi Engineering Journal*. 15, 2019, 60–69. <https://doi.org/10.22153/kej.2019.04.001>
- [105] P.G. shekhi Abadi, M. Irani, L.R. Rad, "Mechanisms of the removal of the metal ions, dyes, and drugs from wastewaters by the electrospun nanofiber membranes," *Journal of the Taiwan Institute of Chemical Engineers*. 143, 2023, 104625. <https://doi.org/10.1016/J.JTICE.2022.104625>
- [106] C. Zhang, Y. Ma, X. Zhang, S. Abdolhosseinzadeh, H. Sheng, W. Lan, A. Pakdel, J. Heier, F. Nüesch, "Two-Dimensional Transition Metal Carbides and Nitrides (MXenes): Synthesis, Properties, and Electrochemical Energy Storage Applications," *Energy and Environmental Materials*. 3, 2020, 29–55. <https://doi.org/10.1002/eam2.12058>
- [107] Y. Gong, X. Xing, Y. Wang, Z. Lv, Y. Zhou, S.-T. Han, "Emerging MXenes for Functional Memories," *Small Science*. 1, 2021, 2100006. <https://doi.org/10.1002/smssc.202100006>
- [108] Z. Ajmal, A. Qadeer, K. Singh, A. Al Mahmud, M.N. Lakhan, H. Pradeep, B. Hussain, A. Ullah, H.R. Khan, E.H. Ibrahim, O.J. Kewate, J. Cui, P. Rosaiah, B. Akkinepally, I. Hussain, W. Shuhang, "A comprehensive review on MXenes for various applications," *Applied Energy*. 397, 2025, 126136. <https://doi.org/10.1016/j.apenergy.2025.126136>
- [109] Ahmaruzzaman, "MXenes and MXene-supported nanocomposites: a novel materials for aqueous environmental remediation.," *RSC Advances*. 12 2022 34766–34789. <https://doi.org/10.1039/d2ra05530a>
- [110] N.N. Safi, B.I. Waisi, "Mathematical and experimental study of the direct contact membrane distillation method for desalination using a hydrophobic electrospun nanofibers membrane," *Iraqi Journal of Chemical and Petroleum Engineering*. 25, 2024, 123–137. <https://doi.org/10.31699/IJCPE.2024.4.12>

- [111] X. Feng, Z. Yu, R. Long, Y. Sun, M. Wang, X. Li, G. Zeng, "Polydopamine intimate contacted two-dimensional/two-dimensional ultrathin nylon basement membrane supported RGO/PDA/MXene composite material for oil-water separation and dye removal," *Separation and Purification Technology*. 247, 2020, 116945. <https://doi.org/10.1016/j.seppur.2020.116945>
- [112] B. Sun, P. Qiu, Z. Liang, Y. Xue, X. Zhang, L. Yang, H. Cui, J. Tian, "The fabrication of 1D/2D CdS nanorod@Ti3C2 MXene composites for good photocatalytic activity of hydrogen generation and ammonia synthesis," *Chemical Engineering Journal*. 406, 2021, 127177. <https://doi.org/10.1016/J.CEJ.2020.127177>
- [113] M.U. Hameed, Z. Amjad, B. Al Alwan, A. El Jery, H.M.A. Hassan, L. Ali, A. Ahmad, S. Khan, S. Raza, "Recent progress in 2D-MXene-based membranes for water treatments; fabrication, properties, and advanced desalination application," *Desalination*. 599, 2025, 118462. <https://doi.org/10.1016/j.desal.2024.118462>
- [114] Soni, M. Ahuja, P.K. Jagtap, V. Chauhan, S.K. Raj, P.P. Sharma, "The Advent of MXene-Based Synthetics and Modification Approaches for Advanced Applications in Wastewater Treatment," *Membranes*. 15, 2025, 1–21. <https://doi.org/10.3390/membranes15120364>
- [115] Z. Zeng, L. Song, L. Ding, H. Wang, "MXene-Based Membranes for Selective Ion Separation," *Membranes*. 16, 2026. <https://doi.org/10.3390/membranes16040146>
- [116] W.K. Al-Musawy, B.I. Waisi, M.H. Al-Furaiji, A. Shehan, M. Jahangiri, "Preparation, Characterization, and Evaluation of Electrospun Copolymer Nanofiber Membranes for Kerosene Removal from Aqueous Solution," *Al-Khwarizmi Engineering Journal*. 22, 2026, 1–13. <https://doi.org/10.22153/kej.2026.12.010>
- [117] K. Abuhasel, M. Kchaou, M. Alquraish, Y. Munusamy, Y.T. Jeng, "Oily Wastewater Treatment: Overview of Conventional and Modern Methods, Challenges, and Future Opportunities," *Water*. 13, 2021. <https://doi.org/10.3390/w13070980>
- [118] Y. Xin, B. Qi, X. Wu, C. Yang, B. Li, "Different types of membrane materials for oil-water separation: Status and challenges," *Colloid and Interface Science Communications*. 59, 2024, 100772. <https://doi.org/10.1016/J.COLCOM.2024.100772>
- [119] Sun, W. Ye, J. Zhang, K. Zheng, "Structure, Properties, and Preparation of MXene and the Application of Its Composites in Supercapacitors," *Inorganics*. 12, 2024. <https://doi.org/10.3390/inorganics12040112>
- [120] A.R. Jabur, L.K. Abbas, S.A. Moosa, "Antibacterial Activity of Electrospun Silver Nitrate /Nylon 6 Polymeric Nanofiber Water Filtration Mats," *Al-Khwarizmi Engineering Journal*. 13, 2017, 84–93. <https://doi.org/10.22153/kej.2017.10.001>
- [121] B. Sarac, S. Yücer, F. Ciftci, "MXenes in microbiology and virology: from pathogen detection to antimicrobial applications," *Nanoscale*. 17, 2025, 9619–9651. <https://doi.org/10.1039/d5nr00477b>
- [122] Y. Liu, X. Chen, J. Sun, N. Xu, Q. Tang, J. Ren, C. Chen, W. Lei, C. Zhang, D. Liu, "Large-scale production of MXenes as nanoknives for antibacterial application," *Nanoscale Advances*. 5, 2023, 6572–6581. <https://doi.org/10.1039/D3NA00744H>
- [123] Basma I. Waisi, "Carbonized Copolymers Nonwoven Nanofibers Composite: Surface Morphology and Fibers Orientation," *Iraqi Journal of Chemical and Petroleum Engineering*. 2, 2019. <https://doi.org/10.31699/IJCPE.2019.2.2>
- [124] S. Raut, J.K. Sahoo, S.K. Sahoo, "Electrospun MXene based functionalized nanofibers for water pollutants adsorption: Potentials, challenges and future perspective," *Journal of Molecular Liquids*. 433, 2025, 127871. <https://doi.org/10.1016/J.MOLLIQ.2025.127871>
- [125] W. Bao, H. Shen, G. Zeng, Y. Zhang, Y. Wang, D. Cui, J. Xia, K. Jing, H. Liu, C. Guo, F. Yu, K. Sun, J. Li, "Engineering the next generation of MXenes: challenges and strategies for scalable production and enhanced performance," *Nanoscale*. 17, 2025, 6204–6265. <https://doi.org/10.1039/D4NR04560B>
- [126] Z.E. AlHadithy, A.A. AbdulRazak, A.M.H.A. Al-Ghaban, Q.F. Alsahy, H. Meskher, R.A. Al-Juboori, K.K. Katibi, D.U. Lawal, "Advancements in Water Treatment with MXene-Enhanced Membranes: A Review of Current Progress and Future Directions," *Water, Air, & Soil Pollution*. 235, 2024, 809. <https://doi.org/10.1007/s11270-024-07628-x>

مركبات MXene: أنواعها، تحضيرها، وتطبيقاتها في معالجة المياه

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الخلاصة

حظيت مواد MXene باهتمام كبير كمادة ثنائية الأبعاد متطورة لمعالجة المياه، إلا أن التقييمات السابقة كانت وصفية في معظمها، ولم تُجرِ مقارنة كافية بين أنواع المواد واستراتيجيات التصنيع والأداء العملي. تهدف هذه الورقة البحثية إلى تقييم الأغشية القائمة على MXene من حيث التركيب، والكيمياء السطحية، وهندسة الطبقات البينية، وتصميم الأغشية، وربط خصائصها بتدفق المياه، وكفاءة الرفض، وقدرة الامتصاص، ومقاومة التلوث، والاستقرار. لا يهدف بحثنا إلى تقديم ملخصات عامة، بل إلى تسليط الضوء على النقاط الحاسمة التي ستوجه استخدام MXene من التجارب المخبرية إلى التطبيقات العملية في تنقية المياه. تتناول هذه المراجعة أيضًا التحديات الرئيسية، بما في ذلك الأكسدة، وإعادة التراص، والتورم، والهشاشة الميكانيكية، وقابلية التوسع، والسلامة البيئية. يوفر هذا التباين بين الثغرات القائمة والتقدم المحرز أساسًا أكثر وضوحًا لاختيار وتصميم أغشية MXene في تطبيقات معالجة المياه المستدامة.

الكلمات الدالة: MXene، Ti_3C_2Tx ، الأغشية، معالجة المياه، الترشيح النانوي.